

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)			Indicative content - alkanes and alkenes both contain C and H - C and H have similar electronegativities - there are no dipoles in either alkanes or alkenes - alkenes have double bonds and are much more reactive - due to high electron density of π bond - caused by p-p sideways overlap - alkanes are saturated whilst alkenes are unsaturated - this makes alkenes susceptible to electrophilic addition - alkanes need light to react - by radical substitution	6			6		
				5-6 marks Explanation of difference in reactivity of alkanes and alkenes in terms of π bond; reference to different reaction mechanisms <i>The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary is used accurately throughout.</i>						
				3-4 marks Explanation of lack of dipoles; reference to alkenes as being reactive and alkanes as being unreactive <i>The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.</i>						
				1-2 marks Some knowledge of the different reactivities of alkanes and alkenes <i>The candidate attempts to link relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.</i>						
				0 marks <i>The candidate does not make any attempt or give an answer worthy of credit.</i>						

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	(b)	(i)		award (1) for either of following - the aldehyde is being oxidised to a carboxylic acid and reduced to an alcohol - the aldehyde gains oxygen forming a carboxylic acid and gains hydrogen forming an alcohol			1	1		
		(ii)		moles of C ₆ H ₅ CHO = $\frac{9.5}{106.06}$ = 0.0896 (1) this should make 0.0448 mol of C ₆ H ₅ CH ₂ OH (1) percentage yield = $\frac{3.39}{4.84} \times 100$ = 70 % (1) accept alternative method moles of C ₆ H ₅ CH ₂ OH = $\frac{3.39}{108.08}$ = 0.0314 (1) formed from 2 × 0.0314 = 0.0628 mol of C ₆ H ₅ CHO (1) mass of C ₆ H ₅ CHO = 6.66 g percentage yield = $\frac{6.66}{9.50} \times 100$ = 70 % (1)		3		3	2	
		(iii)		2C ₆ H ₅ CHO + NaOH → C ₆ H ₅ CH ₂ OH + C ₆ H ₅ COONa			1	1		
				Question 10 total	6	3	2	11	2	0